Proposal for a Swiss AI Initiative Small Project

Project Name

An Open, Multilingual, Multi-hop Knowledge Layer for Apertus

Applicants

(names, institutions, and institutional email addresses; underline scientific lead):

- Joel Barmettler, University of Zurich, joel.barmettler@uzh.ch
- Prof. Dr. Abraham Bernstein, Department of Informatics, University of Zurich, bernstein@ifi.uzh.ch
- Dr. Luca Rossetto, School of Computing, Dublin City University, luca.rossetto@dcu.ie

Request Summary

Cost Dimension	**Amount**
Alps Compute Hours (GPU-h)	6,000 GPU-h
Alps Storage Space (TB)	5 TB
Number of files to store simultaneously	∼300,000

Proposal Description (max. 2 A4 pages)

a. Purpose of the Swiss AI Initiative Small Project

Open language models such as Apertus answer questions about well-known entities from their weights, but fail on the long tail: the many people, places, and works that appear only rarely in their training data [3]. A large share of exactly this knowledge already exists in curated, structured form. Wikidata, a public-domain knowledge graph, describes over a hundred million entities and their relations. Today the only practical way to hand a model this knowledge is to rewrite the relevant part of the graph as text and place it in the prompt, which flattens the graph into prose and spends on the order of a hundred context tokens per entity, on every query.

ConceptFormer grounds the model in the graph itself. A small trainable encoder reads an entity's immediate neighborhood, the nodes and edges around it, as graph structure and compresses it into a few continuous *concept tokens* that the frozen model consumes natively, with no textification step in between. The tokens are computed once per entity and reused across prompts, and the knowledge they carry is attributable: they are built from nothing but the entity's citable Wikidata edges.

Our published work established that the idea works: ConceptFormer won the **best paper award** of its workshop at The Web Conference 2026 [1], and a follow-up study, prepared for submission, confirms it on modern instruction-tuned models. **This proposal funds the next step.** It takes a method validated on one graph, in one language, at one hop, and builds it into a knowledge layer for the Apertus mini models [2] that scales to a million entities, works across languages, and answers multi-hop questions.

b. Current Situation of the Thematic Area

What we have already shown. In two rounds of work we have taken the idea from proposal to a validated method. It is efficient: on questions about entities the encoder never saw in training, eight concept tokens match the accuracy of text baselines that spend five to six times more tokens. It preserves the graph: when we rewire a single edge in the neighborhood and re-encode, the model's answer follows the change at a rate that grows with the token budget, evidence that the structure survives compression and that the model reads it, the causal test recent analyses of graph-token systems call for [7]. And it is general: the same recipe trains on two model families without retuning, and an encoder trained on Wikidata answers questions over graphs it was never trained on.

We also mapped where the method stops today, and each limit becomes a work package below: it has been trained on at most 100,000 entities, with accuracy still rising when we stopped; it works only in the language of the labels it was trained on; and it compresses an entity's immediate neighborhood, not multi-step chains.

How this sits in the field. Two lines of research come close. Text-compression methods (e.g. xRAG [4]) squeeze retrieved passages into few tokens, but do not learn from a graph or test unseen entities and faithfulness. Graph-injection methods (e.g. GraphToken, KBLaM [5, 6]) encode graphs, but re-run per question and need labelled training data. None combines a reusable per-entity encoder, label-free training, and a causal faithfulness test.

c. Activities

This project builds **ConceptFormer v3** on the Apertus mini models (v1.1 Instruct, 0.5B / 1.5B / 4B), with Qwen3 and Gemma-3 as controls. Four work packages, each starting from a result we already have.

WP1 (scale to one million entities). We extend the training corpus from today's 100k to 1M Wikidata entities and complete the data-scaling curve, a step toward covering the full graph. Because accuracy has not saturated, this shows how much more of Wikidata's knowledge the encoder can absorb, and how close concept tokens come to full-text retrieval.

WP2 (multilingual). We train the encoder on target-language graph labels so it works in the languages Apertus was built to serve, and measure whether it answers correctly and in the user's language. This package most directly extends Apertus's multilingual strength.

WP3 (multi-hop). We let the encoder represent a neighbor by its own concept vector, so one shared encoder composes facts across several hops. Early experiments already answer two- and three-hop questions that single-hop tokens cannot; here we build the training data and method to do it at scale.

WP4 (Apertus). We run the full pipeline across the Apertus mini family, producing the first knowledge-injection results on the Swiss open model and the combined system: a multilingual, multi-hop concept layer for Apertus.

Why the cluster. Every model here is small (Apertus-mini scale); what our two consumer GPUs cannot do is generate and process the million-entity and per-language training corpora, and run the resulting sweep. We size the request from a measured baseline: one training run takes six hours on our hardware. The total is about **6,000 GPU-hours over six months**, including a 25% contingency and a named stretch tier; a 2,600-hour fallback still delivers the four core results. We also request 5 TB of storage for corpora, checkpoints, and per-item evaluation outputs.

d. Team

The team has carried this line of work through peer review once already: v1 [1] (Barmettler, Bernstein, Rossetto) won its workshop's best paper award at The Web Conference 2026. J. Barmettler (scientific lead, researcher, UZH) built the current system end to end: data pipeline, training, evaluation, and the multilingual and multi-hop pilots. Prof. A. Bernstein (UZH) brings expertise in knowledge graphs and the Semantic Web; Dr. L. Rossetto (Dublin City University) brings expertise in multimodal retrieval and evaluation. We work against a pre-registered plan with shared dashboards and per-item result dumps, so every claim can be checked independently. [Multi-node/Alps experience to be added.]

e. Expected Outputs and Outcomes

Within six months: (1) a paper reporting the data-scaling law, the multilingual and multi-hop results, and the causal-faithfulness method; (2) open corpora at 300k and 1M entities, multilingual and multi-hop (Wikidata, CC0); (3) open weights for every trained encoder, including the Apertus-mini concept layers; (4) the open, tested pipeline and evaluation harness; (5) the per-item results dataset, so others can re-analyze our findings without a GPU. Together these turn ConceptFormer into a knowledge layer purpose-built for Apertus, and identify the design points worth training at full scale in a follow-on Large Project.

f. Importance for Switzerland, Europe, and beyond

The project fits the Initiative's own model of openness. ConceptFormer's only knowledge source is Wikidata (CC0, public domain), its training uses no proprietary or labelled data, and all code, weights, and corpora are released: the same open-data, open-weights, open-recipe stance on which Apertus was built, and WP2 extends its multilingual reach. The result is a knowledge channel that grounds open models in curated, auditable sources. It is attributable (an entity's tokens are built from nothing but its citable Wikidata edges, and our counterfactual tests show the model uses them), editable (updating the graph and re-encoding the entity takes one encoder pass, no retraining), and compact (at matched accuracy, text costs five to six times more context tokens). For Switzerland and Europe, that supports digital sovereignty: an organization can maintain its own knowledge graph and update what an open model knows, without retraining it or running retrieval infrastructure.

g. Ethics and Regulatory Compliance

Data: Wikidata (CC0) and public benchmarks; no personal data beyond encyclopedic facts about public entities, no human subjects, and no scraping. Training questions are generated by open-weight models and released with their prompts; all artifacts carry explicit licenses. The approach is transparency-enhancing and aligns with EU AI Act expectations; we follow UZH research-integrity and CSCS usage policies and publish all results, including negative ones.

Signature

By submitting this proposal, we confirm that all leading applicants have understood the requirements for participation according to the call document.

Place(s) and Date(s):

Names and signatures of the leading applicants:

References

[1] J. Barmettler, A. Bernstein, L. Rossetto. ConceptFormer: Towards graph-native grounding of large language models via latent concept injection. *WWW Companion '26*, pp. 587–596. Best paper award, hosting workshop. DOI 10.1145/3774905.3794653.

[2] Swiss AI Initiative. Apertus: An open, multilingual language model (mini family: Apertus-v1.1 0.5B/1.5B/4B). `huggingface.co/collections/swiss-ai/apertus-mini`. [Fill technical-report citation / arXiv id.]

[3] Empty shelves or lost keys? Disentangling encoding from recall of long-tail facts in language models. arXiv:2602.14080, 2026.

[4] X. Cheng et al. xRAG: Extreme context compression for retrieval-augmented generation with one token. *NeurIPS*, 2024.

[5] B. Perozzi et al. Let your graph do the talking: Encoding structured data for LLMs. arXiv:2402.05862, 2024.

[6] X. Wang, T. Isazawa, L. Mikaelyan, J. Hensman. KBLaM: Knowledge base augmented language model. *ICLR*, 2025.

[7] Revisiting graph-tokenizing LLMs: A systematic evaluation. arXiv:2605.03514, 2026.